Uni-Spense® III
Dry Ingredient Feeder

Designed for high-volume production. The Uni-Spense III offers the same reliability, accuracy, and performance as our standard Uni-Spense.

ORIGINAL INSTRUCTIONS

SPRAY DYNAMICS
UNI-SPENSE III DRY INGREDIENT FEEDER

CONFIGURATION

This sheet defines the Master Series system for which this manual is supplied. In the event of future needs for parts or service, information on this sheet will assist our Service Technicians and help you get the unit back on-line. Do not remove this sheet from this manual except to make a copy for your service files; then be sure to return this original to the manual.

MASTER SERIES UNI-SPENSE® III DRY INGREDIENT FEEDER

SERIAL NO.: 1242

SALES ORDER: 2020053566

MODEL NAME: UNI-SPENSE III

MOTOR: 1 HP, 208/230-460 VAC, 56C, SS

GEARBOX: 20:1, DODGE, 56C, SS

	TYPE	PART NO.
AUGER TYPE:	MID-RANGE	31743040

POWER REQUIREMENTS: 480 VAC / 3 PH / 60 HZ

CONTROL PANEL: 32095129 SERIAL #1432

SUPPLIED W/ SCARF FEEDER: X YES ____ NO

IMPORTANT INFORMATION FOR ORDERING PARTS:
When ordering parts, be sure to list name of part, part number, quantity required, and Machine Serial Number

PREFACE

SAFETY MEASURES
Throughout this manual there are 4 symbols next to information which should be read and carefully understood before doing any maintenance or operating the machine. The symbols are


Electrical warning

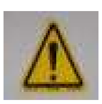

General Warning


Hand Entanglement Rotating Gears


Hand Entanglement Belts/Chains

 System uses compressed air - beware of burst / disconnected air lines.

 DO NOT ATTEMPT TO BYPASS SAFETY INTERLOCKS.

 Use care when replacing cover on hopper – repair any burrs/sharp edges that occur.

 Do not operate seasoning auger without auger tube in place.

  Hazardous Voltages. Before opening any electrical panel, or terminal box, proper Lock-Out/Tag-Out procedures should be followed. Lock-out/Tag-out should also include ANY energy source. These energy sources can include hydraulics, pneumatics, and water. Only competent persons should maintain electrical equipment.

SPRAY DYNAMICS
UNI-SPENSE III DRY INGREDIENT FEEDER

INTRODUCTION

Spray Dynamics Uni-Spense III Dry Ingredient Feeder is designed to accurately meter a volume of product into another piece of equipment. The product may be seasonings, flour, sugar, seeds, or other compatible material.

This equipment is not intended to coat any product outside the scope of the original proposal.

Any variations in the design of this system will void any implied warranty, unless authorized by Spray Dynamics. Any new products to be coated must be assessed for the potential risks and hazards and suitable measures must be taken.

The rate at which product is dispensed is widely adjustable.

Features of the seasoning unit include:

*Consistent metering

*Wide range of adjustments

* Easy cleaning and maintenance

*Exceptional reliability

This Operation and Service Manual contains information required to install, operate, service, troubleshoot, and repair the Uni-Spense III unit. Call our factory for further help and product support.

GENERAL DESCRIPTION

PRODUCT DESCRIPTION

Principle parts and assemblies of the unit are:

*Motor and base assembly

*Transfer tube

*Auger

*Hopper (with vibrator)

*Control Panel

Except for the hopper plug and auger, all product-contact points are constructed of stainless steel. The hopper plug is UHMW, and the auger can be of various materials: including Delrin, UHMW, or Stainless Steel.

Refer to **Principles of Operation** for a detailed description of unit parts and operating principles.

Detailed mechanical assembly and schematic drawings are included in the back section of this manual. Refer to those drawings for specific parts information.

SPECIFICATIONS

The information on the following page gives the general specifications for the Uni-Spense® III unit. A number of features may be especially selected for the specific application. These features may include gear motor ratio, AC power source, transfer tube length, and auger type and length.

The configuration sheet defines the unit for which this manual is furnished. Information on that page may help us to help you if a replacement part is needed or if an operating problem is encountered.
Do not remove that page from this manual.

SPRAY DYNAMICS
UNI-SPENSE III DRY INGREDIENT FEEDER

GENERAL SPECIFICATION

Delivery Rate…………………………….	From 5 to 800 Pounds per hour, depending on motor and auger selection and product characteristics
Auger Types Available………………….	Salting auger (P/N 31013954) Seasoning auger, low range, 1-1/2" pitch (P/N 31013951) Seasoning auger, high range, 2-inch pitch (P/N 31013952)
Motor Gear Ratios ……..………………....	5:1 10:1 15:1 20:1
AC Power Source ……………………….	220/240V AC, 380 V AC, 480 V AC, Three Phase, 50/60Hz.
Fuse ………………………………….….	5A, 10A, 20A, LP-J type, depending on input voltage, refer to electrical drawing.
Mounting……………………………….	Four clamps for 1-1/2"schedule 40 pipe

SPRAY DYNAMICS

UNI-SPENSE III DRY INGREDIENT FEEDER

PRINCIPLES OF OPERATION

GENERAL

This section briefly describes the principle parts of the Uni-Spense III unit and the way in which they work. A clear understanding of this section will help the operator of the equipment, and the maintenance person, to do their job better. The operator must also understand completely the instructions in the **OPERATION** section of this manual.

For detailed parts definitions, and assembly and schematic diagrams, refer to the drawings in the last section of this manual. (Drawing)

ELECTRICAL CIRCUITS

 Hazardous Voltages. Before opening any electrical panel, or terminal box, proper Lock-Out/Tag-Out procedures should be followed. Lock-out/Tag-out should also include ANY energy source. These energy sources can include hydraulics, pneumatics, and water. Only competent persons should maintain electrical equipment.

The cords/cables extending from the control panel are tagged for your convenience. These tags will assist you in properly connecting the power and other external devices together. A minimum of four (4) cables will protrude from the control panel; 1) power, 2) level sensor, 3) plug sensor, 4) Bridge Buster vibrator.

Standard Electrical controls on the Control Panel are:

1. **Start/Stop button:** Press to turn the unit on; push again to turn-off the unit.

2. **Speed Control:** Enter a smaller number to decrease motor speed and product flow rate, and enter a larger number to increase motor speed and product flow rate.

3. The Bridge buster is controlled by a timer within the PLC. A screen in the Operator Interface is used to vary the timing of the hopper vibrating bridge busting basket.

4. **Emergency stop switch:** if pushed, will stop all moving parts. Upon release of the Emergency Stop switch, you must again push the start button to energize the system.

5. **Reset Button:** Resets all safety circuits allowing system to function properly. The system will not work if any of the safety switches are not satisfied. Safety grate and plug must be installed for the safety switch to be satisfied.

MECHANICAL ASSEMBLY

DRIVE MOTOR ASSEMBLY

The drive motor is an AC motor and right angle gearbox. AC power to the motor is furnished by the VFD contained in the Control Panel.

On the output shaft of the gearbox are timing pulleys. These pulleys drive two timing belts. The shorter belt drives a pulley that turns the agitator driver, and the longer belt drives a pulley that turns the main bearing housing which drives the auger. If the auger is installed, it is turned by the revolving main bearing housing.

AUGER AND TRANSFER TUBE

The auger is the screw-like shaft that extends from the main bearing to the end of the transfer tube. Augers are available with different configurations to best suit the characteristics of the specified product.

The auger is held to the main bearing housing by its integral locking hub. The locking hub is secured to the main bearing housing by a simple pin and slot arrangement, which permits the operator to install and remove the auger in a few seconds. The auger is removed by withdrawing the auger from the base.

The auger extends into the transfer tube. The transfer tube is secured, to the unit base by a pair of screw clamps attached to the base.

MECHANICAL ASSEMBLY (Continued)

PRODUCT HOPPER AND FLOW

The product to be dispensed is contained in the stainless steel hopper. A vibrator module attached to the hopper promotes free flow of product. Within the hopper the agitator is turned slowly, to stir the product and further promote product flow downwards into the auger.

The agitator is mounted in flange bearings inserted in opposite sides of the hopper. Each bearing is guarded from the product by a seal designed to increase the wear-life of the flange bearings.

IMPORTANT: When free-flowing products (salt) are being used in the unit, the agitator is normally disengaged. This is done by simply pulling the knob on the agitator drive shaft out approximately on inch, or until the agitator no longer rotates when the drive motor is engaged.

The plug below the hopper and auger permits the assembly to be easily emptied of product and cleaned. This is done by simply opening the two toggle clamps and removing the plug. Note that, if the plug is removed while product is in the hopper, the product will fall from the hopper. There is also a plug sensor. If the plug is removed while the Uni-Spense is running, the sensor will shut off the Uni-Spense motor and thus the auger will stop rotating.

The delivery rate of the product is determined by the Speed Control on the Control Panel in the Operator Interface, which sets the motor speed and, thus, the rate at which product is moved into and through the transfer tube.

SPRAY DYNAMICS

UNI-SPENSE III DRY INGREDIENT FEEDER

INSTALLATION

INSTALLATION PROCEDURES

Use safe lifting practices when installing equipment. Do not stand under equipment being lifted.

The Uni-Spense III Dry Ingredient Feeder may be installed in conjunction with any of several pieces of equipment. The same installation principles apply in either case.

1. Mount the Control Panel on a post using the Control Box Mounting Brackets (supplied with a complete Spray Dynamics tumble drum / Uni-Spense III system) or user-supplied brackets.

*****WARNING*****

Do not apply AC power to the mains at this time.

2. Install the distribution tube on the unit base as follows:

   a. Position the flange of the transfer tube squarely against the flange of the base, then swing in both screw clamps to engage the tube flange, and tighten firmly by hand.

*****WARNING*****

Hand-Tighten screw clamps only. Do not use a wrench or other tool to tighten.

3. Mount the Uni-Spense III unit on (2) 1-1/2-schedule steel pipes. Tighten the clamps on the pipe only enough to ensure that the unit is secure. Do not over tighten.
4. Install the hopper.

INSTALLATION PROCEDURES (Continued)

5. Connect drive motor cable, vibrator cable, and sensor cables to mating cable connectors at the bottom of the Control Panel.

6. Install plug sensor in mounting bracket below Uni-Spense III housing and low level sensor in hopper.

7. Ensure the main disconnect switch is in the **OFF** position.

8. The AC power drop must be connected to the Control Panel. The AC power drop will go directly into the 3-pole fused disconnect switch located in the top right corner of the control panel.

9. Turn on AC power source; generally this is a circuit breaker.

10. Insert the auger in the rear of the base and extend it into the transfer tube until the slots on the auger hub can engage the pins on the rear of the main drive. Then rotate the hub slightly counter-clockwise to engage the pins.

This completes installation procedures. Now check the installation to be sure that the unit is ready for operation.

SPRAY DYNAMICS

UNI-SPENSE III DRY INGREDIENT FEEDER

CHECKOUT PROCEDURES

 All safety circuits must be intact for system to run.

1. Turn on the circuit breaker that feeds power to the unit.
2. Open the front of the Control Panel and measure the supplied voltage. Voltage should be 460-480V AC.

    Only competent persons should maintain electrical equipment.
3. Ensure **all** safety switches are satisfied (Red Plug sensor, and Red grate sensor), then press the **Reset** button.
4. If AC voltage is correct, press the **Uni-Spense ON** button in the Operator Interface. Begin adjusting the **Speed** control upward. The Uni-Spense auger should begin to turn. The vibratory Bridge Buster should also begin to cycle inside the hopper at this point in time. It will cycle based on the settings of the timer. Adjustment of the timer settings may be necessary for each individual spice being applied.
5. Motor should run and auger should turn in Steps 3. If auger does not turn, refer to **Troubleshooting.**
6. Adjust **Speed** control through its range and note that auger speed changes in response to the control.
7. The **Vibrator(s)** will turn off and on based on the settings inside the operator interface. Generally these are set between 1 and 10 seconds off and on. If the vibrators are not required, simply unplug them.

This completes checkout procedures. The unit may be prepared for normal operation (refer to **Preparing for Operation.)**

OPERATION

PREPARING FOR OPERATION

General Procedures

Before starting the run, perform the following steps:

1. If salt or some other easily flowing product is to be dispensed, disconnect the agitator by pulling the knob on the end of the agitator drive shaft out approximately one inch.
2. Close the door on the bottom of the unit and secure with the two toggle latches.
3. Press the **Reset** button on the control panel. This will reset all safety circuits and allow the system to run.
4. Fill the supply hopper with the product to be dispensed.
5. If the product characteristics require vibration to promote flow, make sure the vibrator(s) are plugged in.
6. At the Operator Interface, set Speed Control to the setting that will deliver the desired product flow rate. If this setting has not been determined, perform procedures in Adjusting Flow Rate before starting the run.

If Speed Control settings have been determined, start the run (refer to **Operation.)**

Adjusting Flow Rate

Flow rate (the rate at which the auger must move product into the transfer tube) is determined by motor speed and is adjusted by means of the Speed Control in the Operator Interface. Calibration and adjustment of the Uni-Spense® is done in four easy steps.

Calibration & Adjustment:

Speed control - Represents gear motor speed and serves as a set point for repeatability.

STEP ONE - Determine your application rate based on the feed rate of coated product into the Coating Drum (or across the belt).

-Example-

- Assume you have an uncoated base product rate of 25 lb./min entering the system.
- If you desire a spice application rate of 8% of the base product rate:

25 lbs/min x 0.08 = 2 lbs/min
2 lbs/min is the rate you need to set up the Uni-Spense to run at.

STEP TWO - Develop an application rate curve for each of your spices or salts

****NOTE: ALWAYS DISCONNECT THE AGITATOR WHEN RUNNING SALT!!**

- Set the speed control to 20 and run for one (1) minute using a tray to catch material & then weigh.
- Repeat 3 times. Mark on a graph the average rate for the three catches. Make the graph with axis for Rate (Y-axis) vs. Setting (X-axis)
- Repeat this process at settings of 40, 60, 80, and 100. Mark these on the graph and connect the points. You now have a tool for future reference. Make a graph for each spice.

STEP THREE – Reference the speed setting from the graph for the spice to be applied. Set the speed control and take three (3), one minute samples. Adjust the speed control up or down until you achieve the target application rate. It is a good practice to verify the application rate periodically.

OPERATION

After the flow rate has been adjusted, the unit is ready for a production run.

To start the unit, press the **Uni-Spense START** button. The unit will then dispense the dry product. The operator must monitor the unit to be sure that there is always product in the hopper, and must monitor flow from the transfer tube to detect any visible problem with uniform flow.

The Bridge Buster vibrator is designed to run periodically. Within the Operator Interface, is the Bridge Buster timer. The timer is adjustable to change the **ON/OFF** time of the vibrator. For most spices, an on time for approximately five (5) to ten (10) seconds is sufficient. An off time of between 30 and 45 seconds is sufficient. The operator may, at his/her discretion, turn the Bridge Buster vibrator **OFF** or **ON during operation by simply plugging it in or disconnecting it.**

If any problem is observed, press the **Uni-Spense STOP** button and refer to **TROUBLESHOOTING.**

SANITATION

Machinery should be cleaned and sanitized on a regular schedule to prevent any microbial formations. Each facility should have its own schedule for cleaning and products used for this purpose. ZEP Superior Solutions has several products used for cleaning, degreasing, and sanitizing equipment in the food industry with offices in the United States, France, Italy, Holland, and the United Kingdom.

Cleaning and sanitation should occur daily or more often as your facility may require.

SHUTDOWN AND CLEANUP

When the production run is ended, perform the following steps:

1. Press the Uni-Spense STOP button.
2. Turn the main disconnect switch to the off position.
3. Open toggle latches to allow the plug on the bottom of the unit to open. Any product remaining in the hopper will fall out of the bottom of the unit.
4. Rotate auger hub slightly clockwise until slot clear pins, then pull the auger out of the unit. Any product remaining in the transfer tube will fall out of the bottom of the unit.
5. While supporting the transfer tube, release screw clamps at the tube flange and remove transfer tube.
6. If the components are cleaned, be sure that all are **absolutely dry** before reinstalling.

*****CAUTION*****

Dry product cannot be successfully run through the Uni-Spense III unless all surfaces that contact the product are dry. Certain products may harden following exposure to moisture and cause the auger to break.

MAINTENANCE AND TROUBLESHOOTING

System uses compressed air - check and maintain all connections every 6 months to prevent danger from burst / disconnected lines.

Hazardous voltages inside terminal box/control panel. Lock–out and Tag-out before opening. Only competent persons to maintain electrical equipment.

Use safe lifting practices when installing equipment. Do not stand under equipment being lifted.

Ensure safety switches receive regular maintenance. Check every 6 months for condition and function.

Use care when replacing cover on hopper – repair any burrs/sharp edges that occur.

Lock–out and Tag-out before removing guard to access belts.

Do not operate seasoning auger without transfer tube in place.

ROUTINE CHECKS AND MAINTENANCE

The Uni-Spense III unit requires only minor routine maintenance other than normal cleaning procedures common in the food processing industry.

1. The seals that encase the agitator bearings should be checked weekly for wear. Maintaining good seal condition helps promote longer agitator bearing life (refer to Removal and Replacement Procedures.)
2. The agitator bearings may eventually wear. This is indicated by a visible shaft wobble (refer to Removal and Replacement Procedures.)
3. Grease main bearings monthly using food grade grease. Grease only two pumps.

*****CAUTION*****

Drive motor life may be reduced if motor is run above 90 hertz for an extended period of time

TROUBLESHOOTING

Power is turned on at Control Panel but the gear motor is not heard.

Make sure that the unit is connected to the AC mains and that the associated circuit breaker has not tripped.

Power is at the Control Panel. Drive motor is not heard and neither auger nor agitator driver turns.

1. Make sure that the motor cable is connected to the mating connector on the Control Panel.
2. Verify that the VFD is not faulted. The VFD manual can assist with this and other VFD troubleshooting.
3. Press the Start Uni-Spense button
4. Set Speed Control to 100.
5. Check the VFD (variable frequency drive) inside the control panel. Its digital display should read 60. This is 60 HZ. If there is a display on the VFD, other than 0, check with a multimeter the load leads to the motor. If there is a voltage reading, the motor could be bad. Consult factory for further assistance.
6. Check the VFD parameters. Factory set parameters are listed on the electrical drawing. Refer to VFD manual for checking parameters.
7. If the VFD is determined to be OK, set power switch to the **OFF** position, then manually rotate auger hub. Hub should rotate relatively easy with 5:1 and 10:1 gearboxes. Two hands will be able to rotate the hub with higher-ratio gearboxes. As the hub turns, the gearbox should be heard turning

Agitator turns but auger does not turn.

Ensure that the Drive Hub is engaged to the main bearing.
Check drive train. Replace belts as necessary.

Auger turns but agitator does not turn.

After making sure the disconnect handle is engaged, check the agitator hub for missing or broken drive pin.

Dispensed product leaks through agitator shaft bearings.

Bearings are worn. Replace bearings and seals as required.

REMOVAL AND REPLACEMENT PROCEDURES

REMOVING AND REPLACING MAIN BEARING ASSEMBLY

Main bearing wear is indicated by either: (1) overheating of the bearing during operation, or (2) bearing seizure. To determine bearing seizure, remove the motor assembly and main belt, then try to rotate the bearing manually. If the bearing does not turn freely, the bearing may have seized. If the bearing is seized or overheats during operation, replace it as follows:

1. Remove AC voltage from panel and lock out Control Panel.

2. Remove auger from the Uni-Spense® base.

3. Remove the four socket-head cap screws that hold the vertical housing to the main weldment. Pull the entire vertical housing and motor away from the main weldment. Place on a table.
4. Remove all the button-head socket cap screws that hold the cover to the vertical housing. 12 total
5. Remove the 3 socket-head cap screws that hold the front vertical housing plate to the rear vertical housing plate. These bolts are long.
6. **Using a Spanner Wrench, unscrew the** locking ring at the rear of the main bearing assembly.

7. Pull the main bearing assembly out the rear of the Uni-Spense ® base. The main bearing assembly should slide off the bearing sleeve easily. If the main bearing assembly does not slide off the bearing sleeve easily, either the bearing assembly and/or the bearing sleeve may be damaged. You may also need to replace the bearing sleeve (P/N 31013905). The bearing sleeve is held in place by three (3) 10-32-cap screws on the housing. Be sure that the main timing belt is disengaged from the pulley and is not damaged while removing the bearing assembly.

8. Replace the main bearing assembly (P/N 31013904), pushing it through the rear of the base until it is seated. Be sure to engage the main timing belt on the pulley.

9. Re-install and tighten locking ring removed in Step 6.

10. Install other components in the opposite order of removal. Refer to vertical housing assembly drawing at the end of this manual. (DWG. # UNI III DRIVE EXP)
11. After replacing any internal components on the vertical housing, the 2 piece cover should be re-attached using a 100% silicone sealant that is safe for food contact. Spray Dynamics recommends DAP 100% silicone rubber sealant (clear), as this is FDA compliant under regulation number 21 CFR 177.2600
12. **After re-assembly is complete, install** the auger. Then, grasping the auger hub, manually rotate main bearing assembly, belt, and motor until sure that belt is correctly seated in the pulleys.

This completes replacement procedures for the main bearing assembly.

REMOVAL AND REPLACEMENT PROCEDURES

REMOVING AND REPLACING GEAR-MOTOR ASSEMBLY

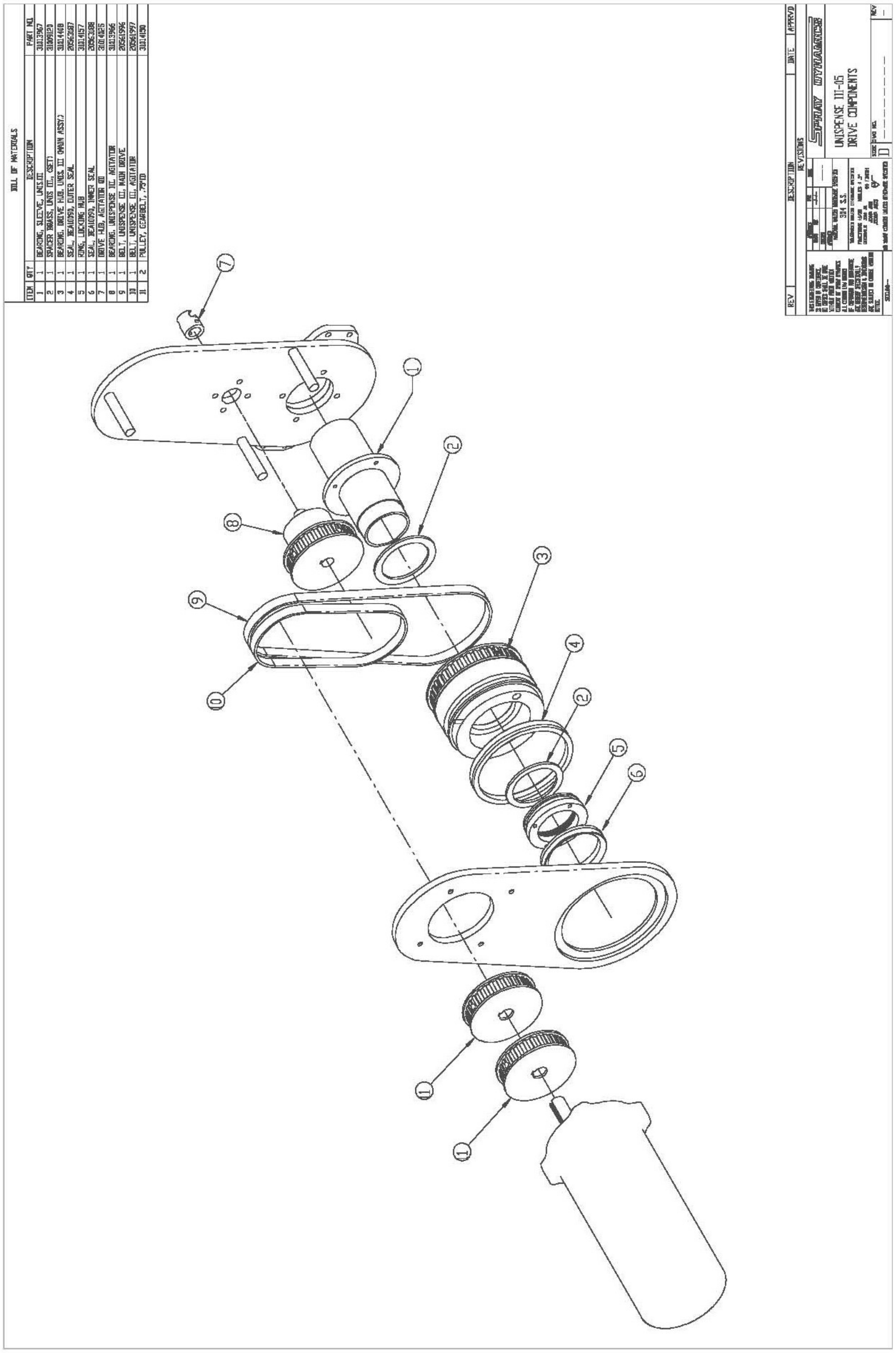
Replace the gear-motor assembly whenever:

> Normal AC voltage is applied to the motor and the output shaft does not turn, or turns with abnormally low speed.

1. Disconnect the motor from control panel.
2. Remove auger.
3. Remove the four socket-head cap screws that hold the vertical housing to the main weldment. Pull the entire vertical housing and motor away from the main weldment. Place on a table.
4. Remove all the button-head socket cap screws that hold's the cover to the vertical housing. 12 total
5. Remove the 3 socket-head cap screws that hold the front vertical housing plate to the rear vertical housing plate. These bolts are long.
6. Remove the set screws that hold the sprocket to the motor shaft. Remove sprocket and belts.
7. Remove the four (4) bolts that hold the gearbox to the vertical cover plate, and then remove the gearbox and motor assembly from the cover plate. Lowering the motor assembly helps to free the output pulley from the belts.
8. Install new gearbox in the opposite order. Refer to vertical housing assembly drawing at the end of this manual. (DWG. # UNI III DRIVE EXP)
9. After replacing gear-motor or any internal components on the vertical housing, the 2 piece cover should be re-attached using a 100% silicone sealant that is safe for food contact. Spray Dynamics recommends DAP 100% silicone rubber sealant (clear), as this is FDA compliant under regulation number 21 CFR 177.2600

This completes replacement procedures for the gear-motor assembly.

SPRAY DYNAMICS
UNI-SPENSE III DRY INGREDIENT FEEDER



SPRAY DYNAMICS
UNI-SPENSE III DRY INGREDIENT FEEDER

RECOMMENDED SPARE PARTS LIST

PART #	QTY	DESCRIPTION
31385096	1	Drive Hub Bearing
31013967	1	Sleeve Bearing
31013966	1	Agitator Drive Bearing
20561992	2*	Bearing 2 Bolt, Plastic
31014918	1	Viton Locking Ring Seal
31743040	1	Auger, Uni III Mid Range Seasoning
31009120	1	Main Bearing Brass Spacer (Set)
20561996	1*	322L075 Main Drive Belt
20561997	1*	210L075 Agitator Belt
31014150	1	Gearbelt Pulley, .75” ID, 30Tooth
20563616	1	AC Motor, 1 HP, 56C, SS
20576326	1	Gearbox, 20:1, SS, Hollow Bore
20563187	1*	Outer Seal for Bearing 31015017
20563188	1*	Inner Seal for Bearing 31015017
20605083	1*	Coil, Soleniod,24VDC,FEST#8030802
20629224	1*	Valve, Solenoid,1/4NPT,VUVS,FEST#575552

*** HIGHLY RECOMMENDED TO HAVE ON HAND**

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